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Predicting the risk of heart disease

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Dedication

We dedicate this project to our parents, who struggled to reach this level of our lives, then to our King and his Crown Prince, who guaranteed us the educational stages and facilitated our path; then to our teachers at all levels of education who worked to educate us and develop our ideas, then to the dean of the faculty and our instructors who opened our horizons to the heights Especially for Dr. Abed Alghaully, our guide and special project officer.

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Chapter 1: Introduction

Abstract

Heart disease remains a leading cause of mortality worldwide. Early detection and prevention are crucial for improving patient outcomes. This project proposes a machine learning-based approach to predict heart disease risk at an early stage. We'll develop and evaluate a machine learning model by leveraging a comprehensive dataset of patient attributes, including demographic information, medical history, and clinical measurements.

The developed model will demonstrate the potential of machine learning techniques to accurately identify individuals at high risk for heart disease, enabling timely interventions and improving overall cardiovascular health.

المستخلص

تظل أمراض القلب أحد الأسباب الرئيسية للوفاة في جميع أنحاء العالم. ويعد الاكتشاف المبكر والوقاية أمراً بالغ الأهمية لتحسين نتائج المرضى. يقترح هذا المشروع نهجاً قائماً على التعلم الآلي للتنبؤ بمخاطر أمراض القلب في مرحلة مبكرة. من خلال الاستفادة من مجموعة بيانات شاملة لسمات المريض، بما في ذلك المعلومات الديموغرافية والتاريخ الطبي والقياسات السريرية، سنطور ونقيم نموذج التعلم الآلي. سيوضح النموذج المطور إمكانات تقنيات التعلم الآلي في تحديد الأفراد المعرضين لخطر الإصابة بأمراض القلب بدقة، مما يتيح التدخلات في الوقت المناسب وتحسين صحة القلب والأوعية الدموية بشكل عام.

1.1 Introduction

Heart disease remains a leading cause of mortality worldwide, posing significant challenges to healthcare systems. In Saudi Arabia, the prevalence of heart-related conditions has been on the rise due to a combination of factors such as lifestyle changes, unhealthy diets, lack of physical activity, and genetic predispositions. Identifying individuals at risk early is critical for effective intervention and prevention, reducing the burden on healthcare systems and improving patient outcomes.

Advancements in technology, particularly in the field of machine learning, have opened new opportunities for healthcare applications. Machine learning algorithms can analyze complex datasets to uncover patterns and relationships that traditional statistical methods may overlook. These algorithms can make highly accurate predictions based on patient data, such as age, cholesterol levels, blood pressure, smoking habits, and other health-related parameters.

This project focuses on developing a machine learning model to predict the risk of heart disease in individuals. By leveraging publicly available datasets and employing various machine learning techniques, this project aims to create a predictive tool that can support healthcare providers in decision-making and early diagnosis. The ultimate goal is to improve patient care by identifying high-risk individuals and enabling timely interventions.

Moreover, this project emphasizes the application of data science in addressing critical health issues in Saudi Arabia. It underscores the importance of using local and global datasets to develop solutions tailored to the specific needs and challenges of the region. Through comprehensive data analysis, feature engineering, and model evaluation, this project aims to produce a reliable and interpretable model that can assist medical professionals.

The integration of machine learning into healthcare systems is not without its challenges. Issues such as data quality, ethical considerations, and model transparency must be addressed to ensure the model's effectiveness and acceptance.

This project will also explore these challenges, aiming to provide a balanced perspective on the use of machine learning in predicting heart disease.

1.2 Project Objectives:

1. **Data Collection and Preprocessing:** To gather a comprehensive dataset containing relevant features related to heart disease (e.g., demographic, clinical, and lifestyle factors) and apply preprocessing techniques to clean and normalize the data for effective analysis.
2. **Model Development and Evaluation:** To develop various machine learning models (such as decision trees, support vector machines, and neural networks) to predict the likelihood of heart disease and evaluate their performance.
3. **Implementation of a Predictive Model:** To create and train a machine learning model that assists healthcare professionals and individuals in detecting heart disease based on user input.

1.3 Problem Statement:

Heart disease is the leading cause of death globally, responsible for approximately 17.9 million deaths each year, according to the World Health Organization (WHO). This accounts for 32% of all global deaths, with a significant proportion occurring due to preventable risk factors. Early detection is crucial for effective intervention and management, as timely diagnosis can lead to improved patient outcomes and reduced healthcare costs.

Despite advancements in medical technology, traditional diagnostic methods often rely on expensive tests and specialized healthcare facilities, which can be inaccessible to many individuals, particularly in low-resource settings. For instance, in the United States, studies indicate that nearly 50% of heart disease cases go undiagnosed until they reach advanced stages, leading to increased morbidity and mortality. The rise of big data and machine learning presents a promising opportunity to transform how heart disease is detected. Machine learning techniques can analyze large datasets, identify patterns, and make predictions based on various risk factors, including age, gender, medical history, lifestyle choices, and clinical measurements. However, there remains a significant gap in the practical application of these techniques for early heart disease detection in real-world settings.

Furthermore, existing predictive models may not be generalizable across diverse populations, limiting their effectiveness and accessibility.

This project aims to develop a robust machine learning model that can accurately predict the likelihood of heart disease based on a comprehensive analysis of diverse datasets. By focusing on key risk factors and employing advanced algorithms, the model will seek to improve early detection capabilities.

1.4 Literature review:

Numerous studies have been conducted on predicting the risk of heart disease. In this section, we will examine a selection of these studies:

1. A study **entitled** “Prediction of Heart Disease Based on Machine Learning Using Jellyfish Optimization Algorithm”. The study **focuses** on using Machine Learning (ML) techniques to predict heart disease, a leading cause of death worldwide. The research leverages the Cleveland heart disease dataset, aiming to create an ML model with optimal predictive performance. The key challenge addressed is the risk of overfitting due to the large number of features (the "curse of dimensionality"). To mitigate this, the study employed the Jellyfish optimization algorithm, which effectively reduced the dataset's dimensionality by selecting the most relevant features.

The study **concluded** that the highest performance was obtained for the SVM classifier model trained on the dataset with the Jellyfish algorithm, with Sensitivity, Specificity, Accuracy, and Area Under Curve of 98.56%, 98.37%, 98.47%, and 94.48%, respectively. The results show that the combination of the Jellyfish optimization algorithm and SVM classifier has the highest performance for use in heart disease prediction. ([Ahmad Ayid Ahmad and Huseyin Polat, July, 2023](#)).

2. Research titled “Effective Heart Disease Prediction Using Machine Learning Techniques: **The aim of this research** is to develop a predictive model for accurately identifying cardiovascular diseases to reduce fatalities associated with these conditions. The study proposes a novel method using k-modes clustering with Huang’s starting point to enhance classification accuracy. Various machine learning models, including Random Forest (RF), Decision Tree (DT), Multilayer Perceptron (MP), and XGBoost (XGB), were employed and optimized using GridSearchCV for hyperparameter tuning. **The models were trained and evaluated on a real-world dataset of 70,000 instances from Kaggle, with an 80:20 data split. Among the models tested, the Multilayer Perceptron with cross-validation outperformed the others, achieving the highest accuracy of 87.28% and an area under the curve (AUC) value of 0.95, highlighting its superior performance in predicting**

cardiovascular diseases. (Chintan M.Bhatt ,Parth Patel And others,6 February 2023);

3. A study entitled “A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method”. The primary aim of this study is to develop an accurate machine learning (ML) model for early heart disease prediction using various feature selection strategies. To achieve this, the researchers applied chi-square, ANOVA, and mutual information methods to identify three feature subsets (SF-1, SF-2, SF-3) and evaluated the performance of ten ML algorithms, including SVM, XGBoost, bagging, decision trees, and random forests. The study also seeks to create an explainable AI framework using SHAP methods to enhance interpretability and build a mobile application capable of providing instant heart disease predictions based on patient symptoms. The evaluation employed both private and public datasets, incorporating SMOTE to address data imbalance and cross-validation methods to ensure robust results.

The findings reveal that the XGBoost algorithm, when applied to the SF-2 feature subset, outperformed other models, achieving 97.57% accuracy, 96.61% sensitivity, 90.48% specificity, 95.00% precision, a 92.68% F1 score, and 98% AUC. This indicates its superior ability to predict heart disease effectively. Additionally, the proposed system offers a practical solution for healthcare providers by facilitating rapid and cost-effective early-stage heart disease detection. (Hosam El-Sofany, Belgacem Bouallegue & Yasser M. Abd El-Latif, October 2024)

4. A study entitled: “An active learning machine technique based prediction of cardiovascular heart disease from UCI-repository database”. The study aimed to utilize machine learning (ML) classifiers to identify critical features that enhance the accuracy of heart disease prediction. By employing various combinations of features and well-known classification algorithms, the research seeks to develop a robust prediction model. Neural network models, including Naïve Bayes and Radial Basis Functions, were implemented to achieve significant prediction accuracy. Additionally, the study explores the effectiveness of Learning Vector Quantization, which demonstrated the highest prediction accuracy among

the state-of-the-art methods for cardiovascular problem prediction.

The study concluded that machine learning techniques hold great potential for improving heart disease prediction accuracy. Neural network models achieved notable results, with Naïve Bayes reaching an accuracy of 94.78% and Radial Basis Functions achieving 90.78%. Learning Vector Quantization stood out as the most effective method with a prediction accuracy of 98.7%. The research highlights the importance of early intervention, personalized medicine, and advancements in technology to reduce the burden of Cardiovascular Heart Disease (CHD). These contributions aim to save lives, improve health outcomes, and optimize healthcare resource allocation, ultimately benefiting both individual patients and society as a whole. (Saravanan Srinivasan and others, August 2023).

1.5 Time Line:

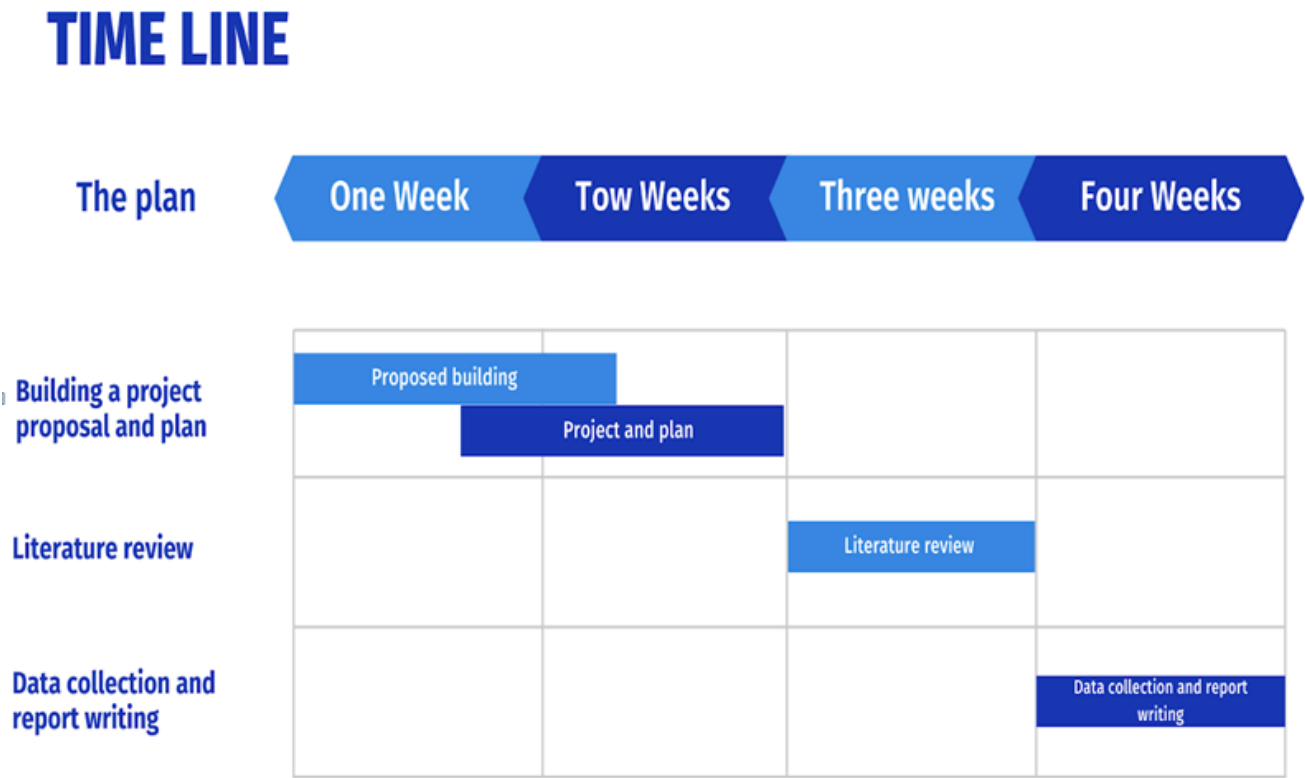


Figure 1.1 timeline

1.6 Software /Hardware Requirements

Software Requirements

1. Programming Language: Python
2. Libraries and Frameworks:
 - NumPy & Pandas: For data manipulation and analysis.
 - Matplotlib & Seaborn: For data visualization and exploratory analysis.
 - Scikit-learn: For implementing machine learning algorithms.

Hardware Requirements

1. Computer.
2. Network Connectivity:
 - To connect to Google colab and download dataset.

Chapter 2: Machine Learning in Healthcare: Applications and Evaluation Metric

2.1 Machine Learning Applications in Healthcare

Machine learning (ML) is revolutionizing the healthcare industry by enabling more accurate diagnoses, personalized treatments, and efficient operations. By leveraging advanced algorithms to analyze vast amounts of medical data, ML is transforming how healthcare professionals deliver care.

One of the most significant applications of ML in healthcare is in the field of medical image analysis. ML algorithms can analyze medical images such as X-rays, MRIs, and CT scans to detect abnormalities and diseases with greater accuracy than human radiologists. This can lead to earlier diagnosis and more effective treatment plans.

Another critical application of ML is in drug discovery and development. By analyzing large datasets of biological and chemical information, ML algorithms can identify potential drug targets and accelerate the development of new medications. This can significantly reduce the time and cost associated with traditional drug discovery processes. Personalized medicine is another area where ML is making a significant impact. By analyzing a patient's genetic information, medical history, and lifestyle factors, ML algorithms can identify the most effective treatment options for individual patients. This can lead to more targeted therapies and improved patient outcomes.

Machine learning algorithms are increasingly used to predict heart disease. By analyzing vast amounts of patient data, including age, blood pressure, cholesterol levels, and other relevant factors, these algorithms can identify patterns and risk factors associated with heart disease. This enables early detection and intervention, potentially saving lives.

Several algorithms are commonly employed, such as logistic regression, decision trees, random forests, and neural networks. These models can be trained on historical data to learn the relationships between different variables and the likelihood of heart disease. Once trained, the models can predict the risk for new patients based on their individual characteristics.

Predictive models can help healthcare providers identify high-risk individuals and recommend lifestyle changes or medical treatments to reduce their risk. Additionally, they can aid in resource allocation and prioritize preventive care efforts. While machine learning shows promise in heart disease prediction, it's crucial to use these models as a tool to support clinical decision-making, not as a replacement for expert medical judgment.

2.2 Classification Algorithms in Machine Learning

Classification algorithms in machine learning are supervised learning techniques designed to sort data into specific categories based on input features. By analyzing labeled training data, these algorithms identify underlying patterns, enabling them to accurately classify new, unseen data.

The following section will explore several well-known algorithms.

2.2.A Random Forest algorithm:

What is random forest?

Random forest is a commonly-used machine learning algorithm, trademarked by Leo Bierman and Adele Cutler, that combines the output of multiple decision trees to reach a single result.

Random forest algorithm

The random forest algorithm is an extension of the bagging method as it utilizes both bagging and feature randomness to create an uncorrelated forest of decision trees.

How it works

Random forest algorithms have three main hyperparameters, which need to be set before training. These include node size, the number of trees, and the number of features sampled.

Random forest applications

- Finance.
- Healthcare.
- E-commerce.

Benefits and challenges of random forest

- Reduced risk of overfitting.
- Provides flexibility.
- Easy to determine feature importance.

2.2.B XGBoost (Extreme Gradient Boosting) Algorithm

Gradient boosting is an ensemble technique that builds models sequentially to improve prediction accuracy. It minimizes loss using gradient descent.

Key Components of Gradient Boosting:

This approach involves several components, including weak learners, a loss function, and a learning rate, which collectively enhance model performance.

Applications of Gradient Boosting Models

- **Healthcare Prognosis and Diagnostics:**

In healthcare, gradient boosting techniques assist in predicting patient outcomes, optimizing treatment plans, and enhancing diagnosis accuracy.

- **Financial Risk Assessment:**

In finance, gradient boosting models are used to predict credit risk, helping institutions assess the likelihood of borrowers defaulting on loans.

- **Customer Churn Prediction:**

Businesses utilize these models to identify customers at risk of leaving, enabling targeted marketing strategies and improving retention rates.

How Does XGBoost Work?

1. **Initialize Predictions:**

- Start with an initial prediction, often the mean value for regression or a uniform probability for classification.

2. **Build Trees Iteratively:**

- Train a decision tree based on the gradient (derivative) of the loss function.
- Add the tree's predictions to the model to improve accuracy.

3. **Optimize Loss:**

- Each tree tries to minimize the error (loss) by focusing on residuals or gradients of the previous step.

4. **Update Predictions:**

- Combine all trees to make the final prediction (weighted sum of tree outputs).

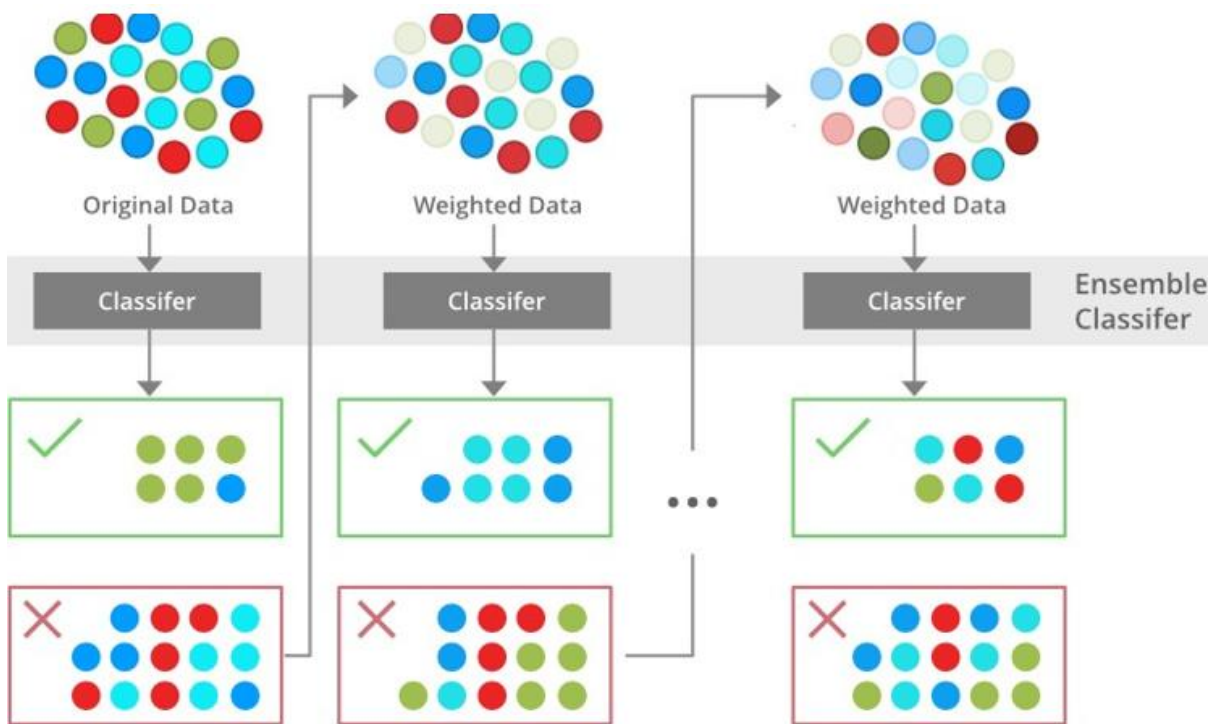


Figure 1.2 XGboost algorithm model

Challenges and Future Directions in Gradient Boosting

- Overfitting Issues in Complex Models:

One challenge is the tendency of gradient boosting to overfit on training data, necessitating techniques like regularization to mitigate this risk.

- Scalability with Large Datasets:

Handling large datasets can be computationally intensive, prompting research into more efficient algorithms that maintain performance while reducing time.

2.2.C Logistic regression algorithm

What is Logistic Regression:

Logistic regression is a supervised machine learning algorithm used for classification tasks where the goal is to predict the probability that an instance belongs to a given class or not.

Types of Logistic Regression:

1. Binomial: In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.
2. Multinomial: In multinomial Logistic regression, there can be 3 or more possible unordered types of the dependent variable, such as “cat”, “dogs”, or “sheep”
3. Ordinal: In ordinal Logistic regression, there can be 3 or more possible ordered types of dependent variables, such as “low”, “Medium”, or “High”.

Advantages of Logistic Regression:

Simplicity

Logistic regression models are mathematically less complex than other machine learning (ML) methods. Therefore, you can implement it even if no one on your team has in-depth machine learning experience.

Speed

Logistic regression models can process large amounts of data at high speed because they require less computational power, such as memory and processing power. This makes it ideal for organisations starting with ML projects to achieve some quick wins.

Flexibility

You can use logistic regression to find answers to questions that have two or more specific outcomes. You can also use it to pre-process data. For example, you can sort data with a large range of values, such as banking, into a finite, smaller range of values using logistic regression. You can then process this smaller dataset using other machine learning techniques for more accurate analysis.

Logistic Regression Application:

manufacture

Manufacturing companies use logistic regression analysis to estimate the likelihood of parts failing in machines. They then plan maintenance schedules based on this estimate to minimize future failures.

Healthcare

Medical researchers plan preventive care and treatment by predicting the likelihood of a patient developing the disease. They use logistic regression models to compare the impact of family history or genes on disease.

Funding

Financial companies are required to analyze financial transactions for fraud and evaluate loan applications and risk insurance applications. These problems are suitable for the logistic regression model because they have separate consequences, such as high risk or low risk and fraud or non-fraudulent.

marketing

Online advertising tools use a logistic regression model to predict users' responses to different words and images and create high-performance ads in which customers participate.

How is Logistic Regression work:

Define the question

Any data analysis begins with a business question. For logistic regression, you must frame the question to get certain results:

Do rainy days affect our monthly sales? (Yes or No)

What kind of credit card activity does a customer perform? (authorized, fraudulent, or potentially fraudulent)

Historical data collection

After selecting the question, you need to determine the data factors involved. You will then collect the previous data for all factors. For example, to answer the first

question above, you can add up the number of rainy days and your monthly sales data for each month in the past three years.

Regression Analysis Model Training

You will process historical data using regression software. The program will manipulate different data points and correlate them mathematically using equations. For example, if the number of rainy days for three months is 3, 5, and 8 and the number of sales in those months is 8, 12, and 18, the regression algorithm will relate the factors to the equation:

$$\text{Number of sales} = 2 * (\text{number of rainy days}) + 2$$

2.2.D Support Vector Machine SVM algorithm

A Support Vector Machine (SVM) is a supervised machine learning algorithm that can be employed for both classification and regression purposes. SVMs are more commonly used in classification problems and as such, this is what we will focus on in this post. SVMs are based on the idea of finding a hyperplane that best divides a dataset into two classes, as shown in Figure1.3.

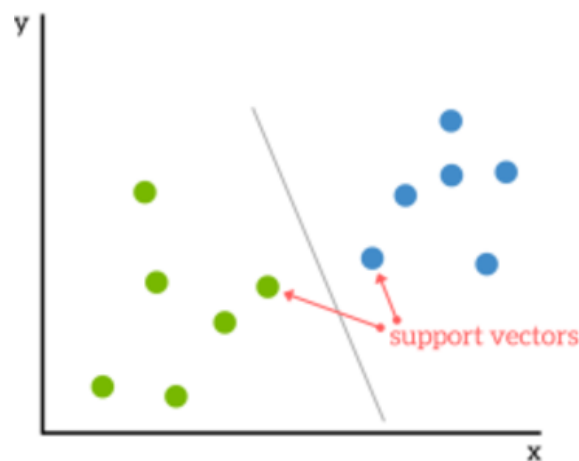


Figure1.3 hyperplane divides a dataset into two classes

Support Vectors

Support vectors are the data points nearest to the hyperplane, the points of a data set that, if removed, would alter the position of the dividing hyperplane. Because of this, they can be considered the critical elements of a data set.

What is a hyperplane?

As a simple example, for a classification task with only two features (like the Figure1.3), you can think of a hyperplane as a line that linearly separates and classifies a set of data.

Intuitively, the further from the hyperplane our data points lie, the more confident we are that they have been correctly classified. We therefore want our data points to be as far away from the hyperplane as possible, while still being on the correct side of it.

So when new testing data is added, whatever side of the hyperplane it lands will decide the class that we assign to it.

How do we find the right hyperplane?

Or, in other words, how do we best segregate the two classes within the data?

The distance between the hyperplane and the nearest data point from either set is known as the margin. The goal is to choose a hyperplane with the greatest possible margin between the hyperplane and any point within the training set, giving a greater chance of new data being classified correctly.

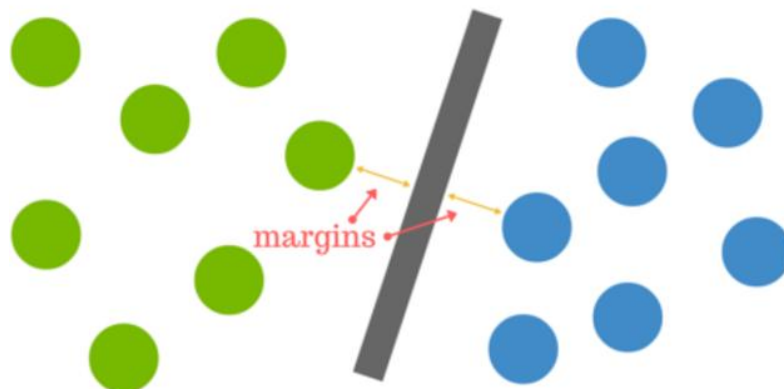


Figure 1.4 a hyperplane with the greatest possible margin

But what happens when there is no clear hyperplane?

This is where it can get tricky. Data is rarely ever as clean as our simple example above. A dataset will often look more like the jumbled balls below which represent a linearly non-separable dataset.

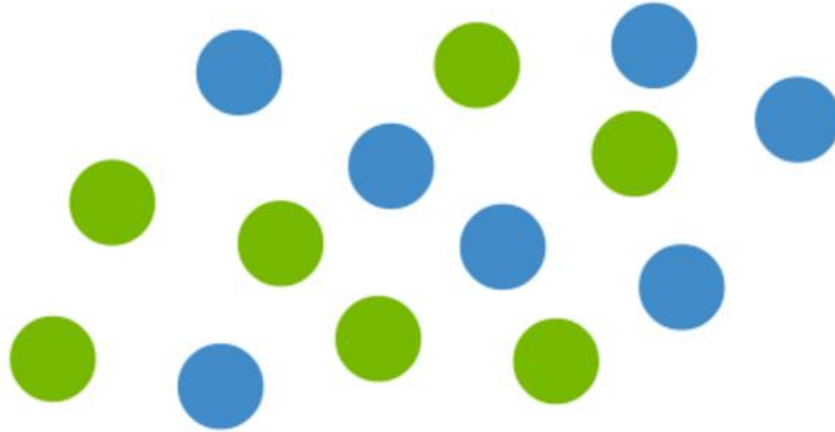


Figure 1.5 linearly non-separable dataset.

In order to classify a dataset like the one above it's necessary to move away from a 2d view of the data to a 3d view. Explaining this is easiest with another simplified example. Imagine that our two sets of colored balls above are sitting on a sheet and this sheet is lifted suddenly, launching the balls into the air. While the balls are up in the air, you use the sheet to separate them. This 'lifting' of the balls represents the mapping of data into a higher dimension. This is known as kernelling.

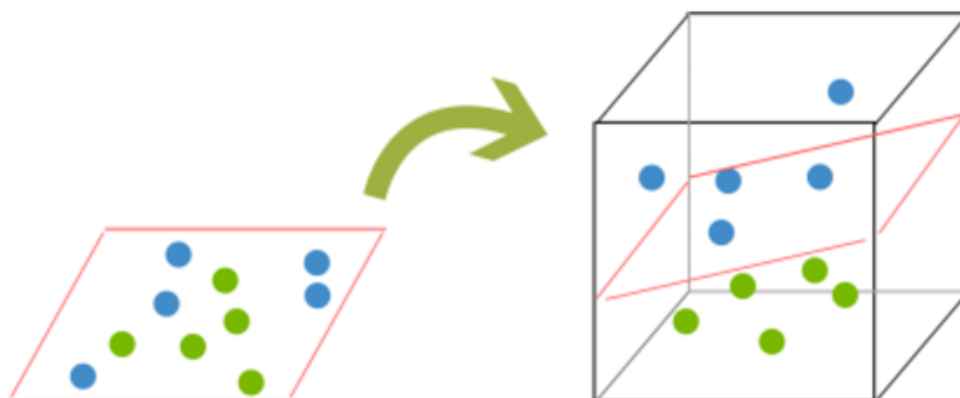


Figure 1.6 kernelling

Advantages

- Accuracy
- Works well on smaller cleaner datasets
- It can be more efficient because it uses a subset of training points

Challenges

- Isn't suited to larger datasets as the training time with SVMs can be high
- Less effective on noisier datasets with overlapping classes

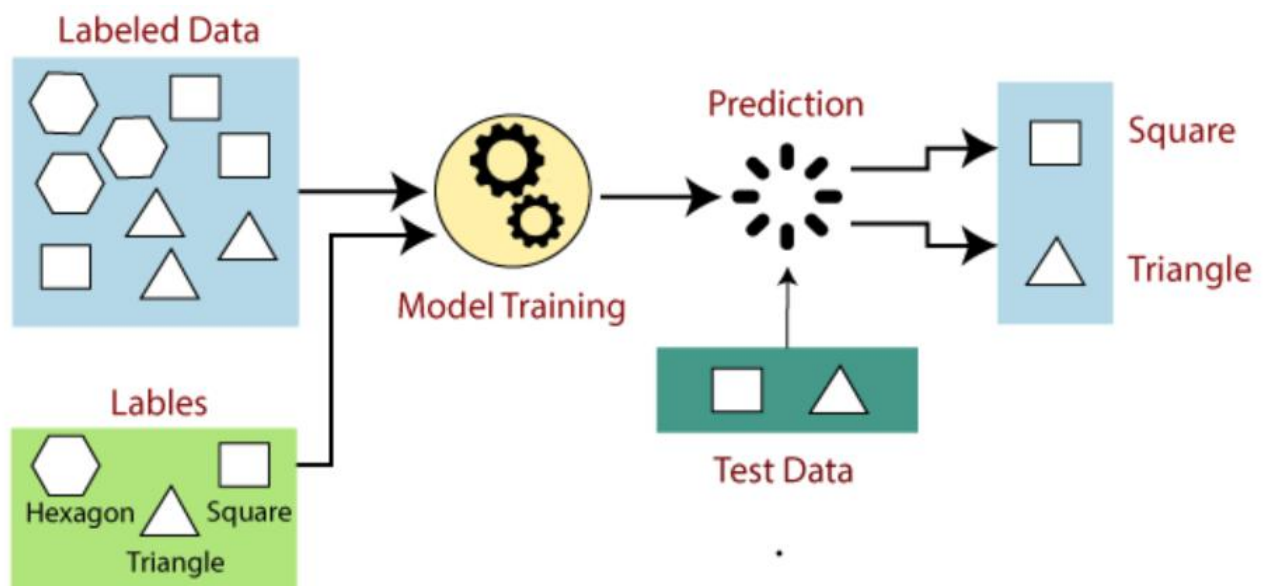


Figure1.7 Supervised Machine Learning Process

2.2.3 Performance Metrics for Classification

In a classification problem, the category or classes of data is identified based on training data. The model learns from the given dataset and then classifies the new

data into classes or groups based on the training. It predicts class labels as the output, such as *Yes or No, 0 or 1, Spam or Not Spam*, etc.

To evaluate the performance of a classification model, different metrics are used, and some of them are as follows:

- **Accuracy**
- **Confusion Matrix**
- **Precision**
- **Recall**
- **F-Score**

1 Accuracy

The accuracy metric is one of the simplest Classification metrics to implement, and it can be determined as the number of correct predictions to the total number of predictions.

It can be formulated as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total number of Predictions}} \quad (1)$$

When to Use Accuracy?

It is good to use the Accuracy metric when the target variable classes in data are approximately balanced.

When not to use Accuracy?

It is recommended not to use the Accuracy measure when the target variable majorly belongs to one class. For example, Suppose there is a model for a disease prediction in which, out of 100 people, only five people have a disease, and 95 people don't have one. In this case, if our model predicts every person with no disease (which means a bad prediction), the Accuracy measure will be 95%, which is not correct.

2 Confusion Matrix

A confusion matrix is a tabular representation of prediction outcomes of any binary classifier, which is used to describe the performance of the classification model on a set of test data when true values are known.

A typical confusion matrix for a binary classifier looks like the below image(However, it can be extended to use for classifiers with more than two classes).

		Predicted	
		Positive	Negative
Actual	Positive	Ture positive	False negative
	Negative	False positive	Ture negative

In general, the table is divided into four terminologies, which are as follows:

1. **True Positive(TP):** In this case, the prediction outcome is true, and it is true in reality, also.
2. **True Negative(TN):** in this case, the prediction outcome is false, and it is false in reality, also.
3. **False Positive(FP):** In this case, prediction outcomes are true, but they are false in actuality.
4. **False Negative(FN):** In this case, predictions are false, and they are true in actuality.

3 Precision

The precision metric is used to overcome the limitation of Accuracy. The precision determines the proportion of positive prediction that was actually correct. It can be calculated as the True Positive or predictions that are actually true to the total positive predictions (True Positive and False Positive).

IV. Recall or Sensitivity

It is also similar to the Precision metric; however, it aims to calculate the proportion of actual positive that was identified incorrectly. It can be calculated as

True Positive or predictions that are actually true to the total number of positives, either correctly predicted as positive or incorrectly predicted as negative (true Positive and false negative).

The formula for calculating Recall is given below:

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

When to use Precision and Recall?

if we want to minimize the **false negative**, then, **Recall** should be as near to 100%, and if we want to minimize the **false positive**, then **precision** should be close to 100% as possible.

In other words, *if we maximize precision, it will minimize the FP errors, and if we maximize recall, it will minimize the FN error.*

4 V. F-Scores

F-score or F1 Score is a metric to evaluate a binary classification model on the basis of predictions that are made for the positive class. It is calculated with the help of Precision and Recall. It is a type of single score that represents both Precision and Recall. So, ***the F1 Score can be calculated as the harmonic mean of both precision and Recall, assigning equal weight to each of them.***

The formula for calculating the F1 score is given below:

$$F1 - score = 2 * \frac{precision*recall}{precision+recall} \quad (3)$$

When to use F-Score?

As F-score make use of both precision and recall, so it should be used if both of them are important for evaluation, but one (precision or recall) is slightly more important to consider than the other. For example, when False negatives are comparatively more important than false positives, or vice versa.

Conclusion

In conclusion, predicting the risk of heart disease using machine learning offers significant potential for early detection and intervention, potentially saving lives. Through the use of various classification algorithms such as decision trees, logistic regression, and support vector machines, we can accurately assess the likelihood of heart disease based on a variety of health metrics. The effectiveness of these models can be evaluated using performance metrics like accuracy, precision, recall, and F1 score, ensuring that the predictions are both reliable and actionable. Machine learning techniques not only enhance our ability to process complex data but also provide valuable insights that can guide medical practitioners in decision-making. As the field continues to evolve, further advancements in algorithm development and data quality will likely improve the precision of heart disease risk predictions, contributing to better healthcare outcomes.

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